

Mechanics of Solids (3130608)

Unit-4

Simple stresses & strains



Prof. Krunal J Dhandha

 krunal.dhandha@darshan.ac.in

 99987 00503

Syllabus – Module 3 Description



GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering
Subject Code: 3130608

MODULE 4

7	Simple stresses & strains Basics of stress and strain: 3-D state of stress (Concept only) Normal/axial stresses: Tensile & compressive Tangential Stresses :Shear and complementary shear Strains: Linear, shear, lateral, thermal and volumetric. Hooke's law, Elastic Constants: Modulus of elasticity, Poisson's ratio, Modulus of rigidity and bulk modulus and relations between them with derivation. Application of normal stress & strains: Homogeneous and composite bars having uniform & stepped sections subjected to axial loads and thermal loads, analysis of homogeneous prismatic bars under multidirectional stresses	10	20
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Topics covered in Presentation

- ▶ Stress – Strain with hook's law
- ▶ Deformation of a body due to force action on it
- ▶ Numerical of simple stress-strain
- ▶ Deformation of a body due to self weight
- ▶ Principle of Superposition
- ▶ Numerical
- ▶ Thermal stress & strain & its numerical
- ▶ Important Definition
- ▶ Volumetric strain of a rectangular body subjected to axial load
- ▶ Volumetric strain of a rectangular body subjected to three mutually perpendicular forces
- ▶ Relation between :- (1) Young's Modulus (E) (2) Bulk Modulus (K) (3) Poisson's ratio (μ)
- ▶ Shear stress, shear strain & Modulus of rigidity or shear modulus
- ▶ Relation between : (1) Young's modulus (E), (2) Modulus of rigidity (G) & (3) Poisson's ratio (μ or $1/m$)

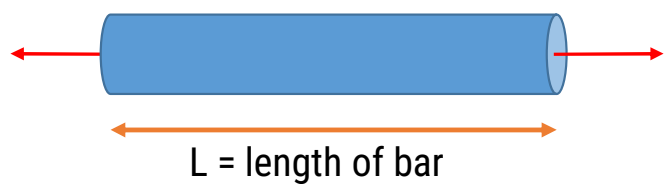
Definition: Stress, Strain & Hook's Law

- ▶ Stress :- It is defined as the resistance developed by the body against applied load, and is defined as load per unit area.

Mathematically, $Stress(\sigma) = \frac{Load (P)}{Cross - sectional\ area (A)} \Rightarrow \sigma = \frac{P}{A}$ N/mm² or kN/m²

- ▶ Strain :- It is defined as ratio of change in dimension to original dimension of specimen

$$Strain(\varepsilon) = \frac{\delta D}{D} \quad (\text{Unit less quantity}) \text{ where, } D = \text{dimension}$$



$$\varepsilon = \frac{\delta l}{l} \quad \text{Where, } \delta l = \text{Change in length \& } l = \text{Original length}$$

- ▶ Hook's Law :- When a material is loaded, within its elastic limit, the stress is proportional to strain

$$\sigma \propto \varepsilon \Rightarrow \sigma = E \times \varepsilon \quad \therefore E = \frac{Stress(\sigma)}{Strain(\varepsilon)}$$

It may be noted that hook's law equally holds good for tension as well as compression

Where, E = Young's modulus **or** modulus of Elasticity (N/mm²)

Deformation of a body due to force action on it

- ▶ Consider a bar element, which is subjected to a tensile force as shown in figure

Where,

P = Load acting on the body

l = Length of the body

A = Cross- sectional area of the body

δl = Change in length



- ▶ As we know that stress and strain

$$\sigma = \frac{P}{A} \quad \varepsilon = \frac{\delta l}{l}$$

- ▶ Now, modulus of elasticity & according to hook's law

$$E = \frac{\sigma}{\varepsilon} \Rightarrow E = \frac{P}{A \times \varepsilon} \Rightarrow E = \frac{P \times l}{A \times \delta l}$$

- ▶ Now, deformation,

$$\delta l = \frac{Pl}{AE}$$

Ex - 1

A steel rod 1m long and 20mm x 20mm in cross-section is subjected to a tensile force of 40kN. Determine the elongation of the rod, if modulus of elasticity for the rod material is 200 GPa.

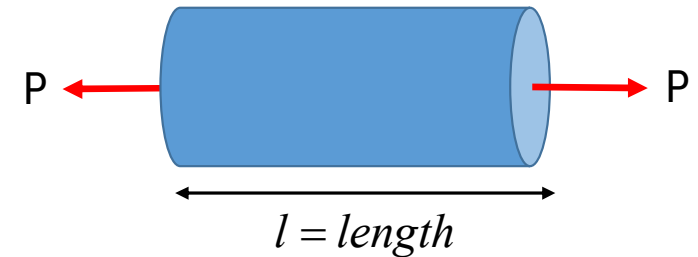
Given Data,

Length (l) = 1m = 1×10^3 mm

Cross-sectional area (A) = $20 \times 20 = 400$ mm²

Tensile force (P) = 40kN = 40×10^3 kN

Modulus of Elasticity (E) = 200GPa = 200×10^3 N/mm²



We know that elongation of the rod is

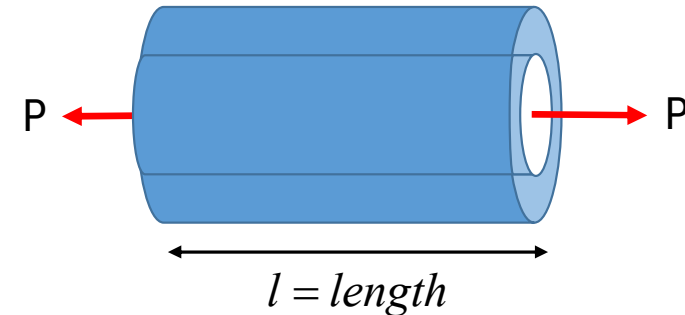
$$\delta l = \frac{Pl}{AE} \quad \Rightarrow \quad \delta l = \frac{40 \times 10^3 \times 1 \times 10^3}{400 \times 200 \times 10^3} = 0.5 \text{ mm}$$

Ex - 2

A hollow cylinder 2m long has an outside diameter of 50mm and inside diameter of 30mm. If the cylinder is carrying a load of 25kN. Find the stress in the cylinder, also find the deformation of the cylinder, if the value of modulus of elasticity for the cylinder material is 100GPa.

Given Data,

Length (l)	= 2m = 2×10^3 mm
Outside diameter (D)	= 50mm
Inside diameter (d)	= 30mm
Applied force (P)	= 25kN = 25×10^3 N
Modulus of Elasticity (E)	= 100GPa = 100×10^3 N/mm ²



Area of hollow cylinder

$$A = \frac{\pi}{4}(D^2 - d^2) = \frac{\pi}{4}(50^2 - 30^2) = 1257 \text{ mm}^2$$

Stress in cylinder

$$\sigma = \frac{P}{A} = \frac{25 \times 10^3}{1257} = 19.9 \text{ N / mm}^2$$

Deformation of the cylinder

$$\delta l = \frac{Pl}{AE} = \frac{25 \times 10^3 \times 2 \times 10^3}{1257 \times 100 \times 10^3} = 0.4 \text{ mm}$$

Deformation of a body due to self weight

► Consider a bar AB hanging freely under its own weight as shown in figure

Let's l = Length of the bar

A = Cross-sectional area of the bar

E = Young's modulus for the bar

w = Specific weight of the bar

Now, consider a small section dx of the bar at a distance x from B

We know that weight of the bar for a length x

$$P = w \times A \times x$$

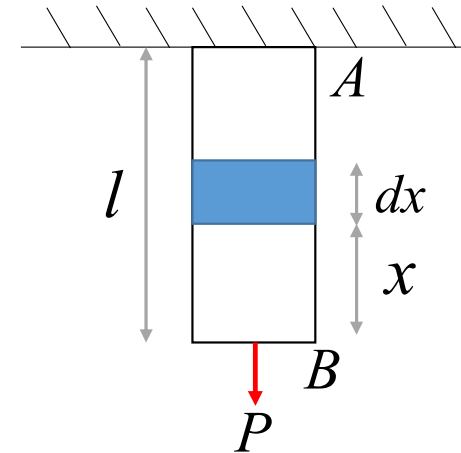
Elongation of the small section of the bar, due to weight of the bar for a small section of length x ,

$$= \frac{Pl}{AE} = \frac{(wAx)dx}{AE} = \frac{(wx)dx}{AE}$$

Total elongation of the bar may be found out by integration the above equation between zero & length (l)

There total elongation

$$\delta l = \int_0^l \frac{w \cdot x \cdot dx}{E} = \frac{w}{E} \int_0^l x \cdot dx = \frac{w}{E} \left[\frac{x^2}{2} \right]_0^l \Rightarrow \delta l = \frac{wl^2}{2E} = \frac{w \times A \times l \times l}{2 \times A \times E} = \frac{Wl}{2AE} \quad (W = wAl = \text{Total weight})$$



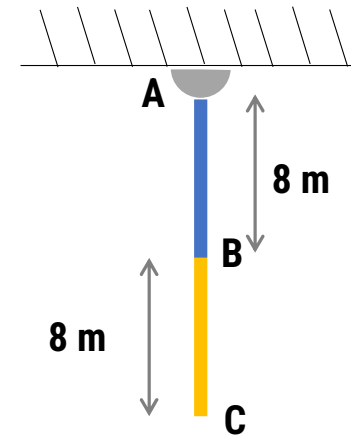
Ex - 3

A steel wire ABC 16m long having cross-sectional area of 4mm² weight 20N as shown in figure. If the modulus of elasticity for the wire material is 200GPa, find deflections at C and B.

Given Data,	Length (l)	= 16m = 16 x 10 ³ mm
	Cross-sectional area (A)	= 4mm ²
	Weight of wire ABC (w)	= 20N
	Modulus of Elasticity (E)	= 200 x 10 ³ N/mm ²

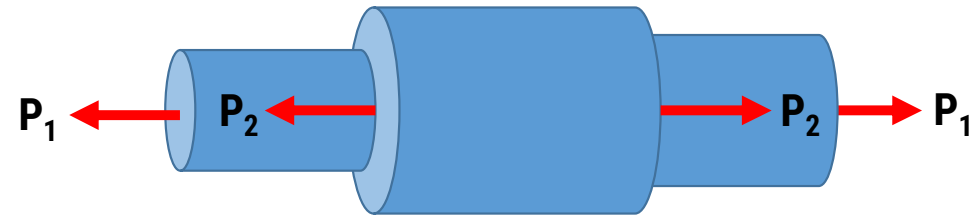
Deflection at C due to self weight AC

$$\delta l = \frac{Wl}{2AE} = \frac{20 \times 16 \times 10^3}{2 \times 4 \times 2 \times 10^3} \Rightarrow \delta l = 0.2mm$$



Principle of Superposition

- ▶ Sometimes, a body is subjected to a numbers of forces acting on its outer edges as well as at some other sections, along the length of the body. In such case, the forces are split up & their effects are considered on individual sections.
- ▶ The resulting deformation, of the body, is equal to the algebraic sum of the deformations of the individual sections. Such a principle of finding out the resultant deformation, is called the principle of superposition.



The relation for the resulting deformation may be modified as

$$\delta l = \frac{Pl}{AE} = \frac{1}{AE} (P_1 l_1 + P_2 l_2 + P_3 l_3 + \dots)$$

Where

P_1 = Force acting on section 1,

l_1 = Length of section 1,

$P_2 l_2$ = Corresponding value of section 2 & so on

But many a times, there are many variations in sectional properties, & material properties.

$$\delta l = \delta l_1 + \delta l_2 + \delta l_3 + \dots + \delta l_n \quad \text{Where}$$

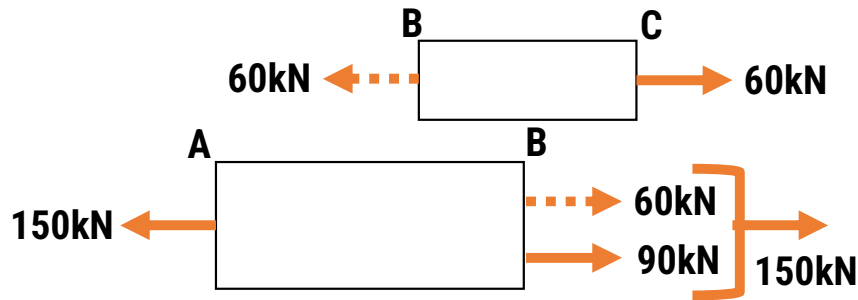
$$\delta l = \frac{P_i l_i}{A_i E_i} \quad \text{i} = 1 \text{ to } n$$

Ex - 4

Calculate total deformation in stepped bar shown in figure. Take modulus of elasticity for both members as $1.2 \times 10^5 \text{ N/mm}^2$.

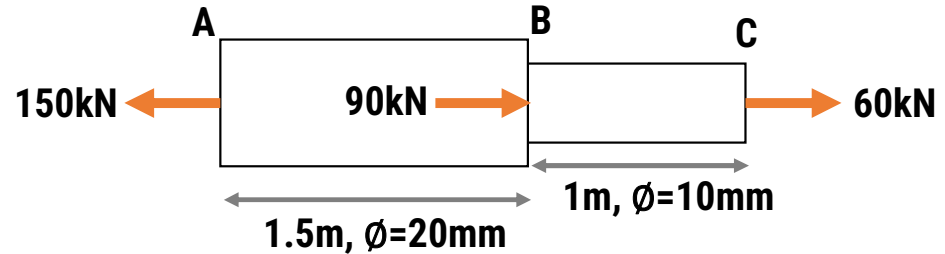
Given Data, $d_{AB} = 20\text{mm}$ $d_{BC} = 10\text{mm}$
 $l_{AB} = 1.5\text{m} = 1.5 \times 10^3\text{mm}$
 $l_{BC} = 1.0\text{m} = 1 \times 10^3\text{mm}$
 $E = 1.2 \times 10^5 \text{ N/mm}^2$

Forces in each members



From free body diagram

$P_{AB} = 150 \text{ kN}$ (Tension), Tensile = +ve
 $P_{BC} = 60 \text{ kN}$ (Tension) Compressive = -ve



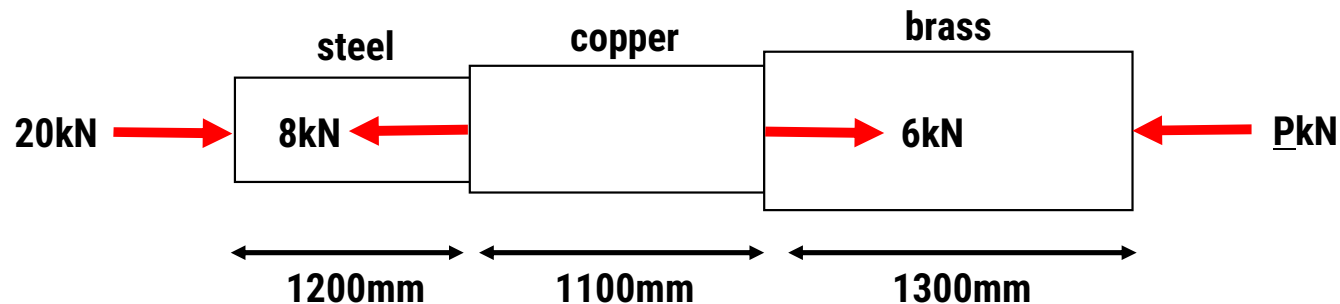
Axial deformation

$$\delta l_{AB} = \frac{P_{AB} l_{AB}}{A_{AB} E_{AB}} = \frac{150 \times 10^3 \times 1500}{(\pi / 4) \times 20^2 \times 1.2 \times 10^5} = +5.968\text{mm} \quad (\text{extension})$$

$$\delta l_{BC} = \frac{P_{BC} l_{BC}}{A_{BC} E_{BC}} = \frac{150 \times 10^3 \times 1000}{(\pi / 4) \times 20^2 \times 1.2 \times 10^5} = +6.366\text{mm} \quad (\text{extension})$$

$$\delta l = \delta l_{AB} + \delta l_{BC} = +12.334\text{mm} \quad (\text{extension})$$

Ex - 5:- A stepped bar made of steel, copper & brass is under axial force as shown in figure & is in equilibrium. The diameter of steel is 12mm, diameter of copper is 16mm & the diameter of brass is 20mm. Determine (i) Magnitude of unknown force "P" (ii) Stress in each material & (iii) Total change in length of the bar. Take Esteel = 200 GPa, E_{copper} = 100 GPa and E_{brass} = 80 GPa

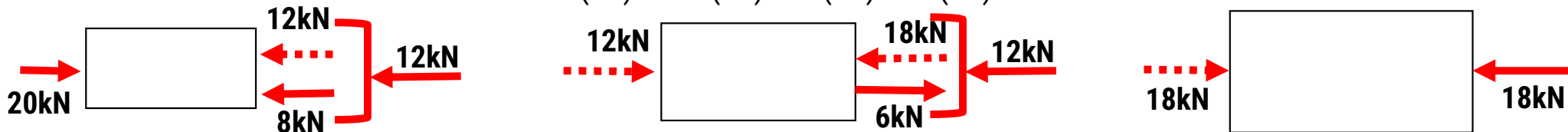


Given Data,

$$\begin{aligned}
 l_s &= 1200 \text{ mm} & d_s &= 12 \text{ mm} & A_s &= (\pi / 4) d_s^2 = 113.1 \text{ mm}^2 & E_s &= 200 \text{ GPa} = 200 \times 10^3 \text{ N/mm}^2 \\
 l_c &= 1100 \text{ mm} & d_c &= 16 \text{ mm} & A_c &= (\pi / 4) d_c^2 = 201.06 \text{ mm}^2 & E_c &= 100 \text{ GPa} = 100 \times 10^3 \text{ N/mm}^2 \\
 l_b &= 1300 \text{ mm} & d_b &= 20 \text{ mm} & A_b &= (\pi / 4) d_b^2 = 314.15 \text{ mm}^2 & E_b &= 80 \text{ GPa} = 80 \times 10^3 \text{ N/mm}^2
 \end{aligned}$$

We know that for equilibrium $\Sigma F_x = 0$. i.e. algebraic sum of all forces in X – direction should be zero. Taking force acting in positive X – direction as positive and negative X – direction as negative.

$$\Sigma F_x = + 20 (\rightarrow) - 08 (\leftarrow) + 6 (\rightarrow) - P (\leftarrow) = 0$$



$$\sigma_s = -\frac{P_s}{A_s} = -\frac{20 \times 10^3}{113.09} = -176.85 \text{ N/mm}^2 \text{ (Compressive)}$$

$$\sigma_c = -\frac{P_c}{A_c} = -\frac{12 \times 10^3}{201.06} = -59.68 \text{ N/mm}^2 \text{ (Compressive)}$$

$$\sigma_b = -\frac{P_b}{A_b} = -\frac{18 \times 10^3}{314.15} = -57.29 \text{ N/mm}^2 \text{ (Compressive)}$$

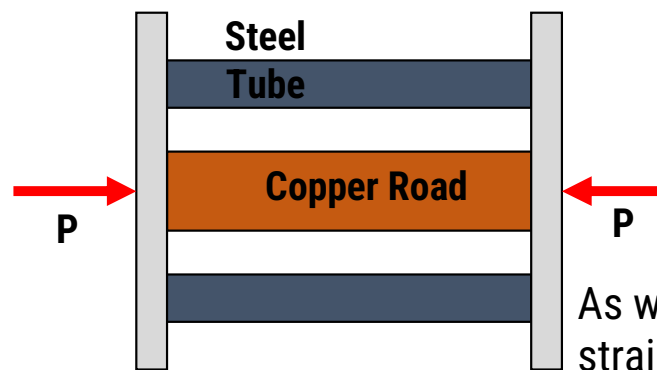
$$\delta l_s = -\frac{P_s l_s}{A_s E_s} = -\frac{20 \times 10^3 \times 1200}{113.09 \times 2 \times 10^5} = -1.06 \text{ mm (Contraction)}$$

$$\delta l_c = -\frac{P_c l_c}{A_c E_c} = -\frac{12 \times 10^3 \times 1100}{201.06 \times 100 \times 10^3} = -0.65 \text{ mm (Contraction)}$$

$$\delta l_b = -\frac{P_b l_b}{A_b E_b} = -\frac{18 \times 10^3 \times 1300}{314.15 \times 80 \times 10^3} = -0.93 \text{ mm (Contraction)}$$

$$\delta l = \delta l_s + \delta l_c + \delta l_b = -2.64 \text{ (Contraction)}$$

Ex – 6:- A copper rod of 20mm dia. is placed inside the steel tube of external dia. 25 mm & internal dia. 22 mm . The tube & rod are 500 mm long & are rigidly connected together at the ends. Find the value of the external load P carried by the assembly if the stress in steel is limited to 150 N/mm^2 . Take $E_s = 200 \text{ kN/mm}^2$ & $E_c = 100 \text{ kN/mm}^2$



Given Data,

$$E_s = 200 \text{ kN/mm}^2, \quad d_c = 20 \text{ mm}, \quad D_s = 25 \text{ mm}, \quad d_s = 22 \text{ mm}$$

$$E_c = 100 \text{ kN/mm}^2, \quad A_c = (\pi/4)(20^2) = 314.15 \text{ mm}^2,$$

$$\sigma_s = 150 \text{ N/mm}^2, \quad A_s = (\pi/4)(25^2 - 22^2) = 110.74 \text{ mm}^2$$

As we know that the given bar is a composite section of steel of steel & copper. So, strains developed in both are equal

Here,

$$\epsilon_s \text{ (Strain in steel)} = \epsilon_c \text{ (Strain in copper)}$$

$$\frac{\sigma_s}{E_s} = \frac{\sigma_c}{E_c} \Rightarrow \sigma_s = \frac{\sigma_c E_s}{E_c}$$

$$150 = \frac{2 \times 10^5 \sigma_c}{1 \times 10^5} = 2\sigma_c \Rightarrow \sigma_c = 75 \text{ N/mm}^2$$

Here, applied force is taken or resist by both material and both of having equal in length

$$\begin{aligned} P &= P_s + P_c = \sigma_s A_s + \sigma_c A_c \\ &= 150 \times 110.74 + 75 \times 314.15 \\ &= 40172.25 \text{ N} \\ &= 40.172 \text{ kN} \end{aligned}$$

Thermal stress & strain

Thermal stress

- ▶ With increase in temperature material expands & with decrease in temperature it contract. If this free expanse or contraction is prevented, stress will be generated in a material. It is called thermal stress.
- ▶ Free deformation of material due to change in temperature

$$\delta l = l \times \alpha \times t \quad \text{Where, } \alpha = \text{co-efficient of thermal expansion} \quad \& \quad t = \text{change in temperature}$$

We know that

$$\delta l = \frac{Pl}{AE} \Rightarrow \delta l = \frac{l}{E} \left(\frac{P}{A} \right) \Rightarrow \delta l = \frac{l}{E} (\sigma) \Rightarrow l \times \alpha \times t = \frac{l}{E} (\sigma) \Rightarrow \sigma = E \times \alpha \times t$$

Thermal strain

Thermal strain = (change in length due to change in temperature) / (Original length)

$$\varepsilon = \frac{\delta l}{l} \Rightarrow \varepsilon = \frac{l \times \alpha \times t}{l} \Rightarrow \varepsilon = \alpha \times t$$

Ex – 7:- A steel rails is 10m long & is laid at temperature of 20°C. The maximum temperature is expected is 500°C, Estimate the minimum gape between two rails to be left so that thermal stress do not developed, The coefficient of linear expansion $\alpha_{\text{steel}} = 12 \times 10^{-6}$ per °C per unitlength

Give data:

$$l = 10m$$

$$\alpha = 12 \times 10^{-6} \text{ per } ^\circ C$$

$$t = 50^\circ - 20^\circ = 30^\circ$$

$$\delta l = l \times \alpha \times t = 10000 \times 12 \times 10^{-6} \times 30 = 3.6mm \quad \text{Minimum gap between two rails}$$

Important Definition

Linear Strain (ϵ):- Deformation per unit dimension is called strain. Strain measured along the direction of load is called linear strain.

$$\text{strain}(\epsilon) = \frac{\text{Change in length}}{\text{Original length}} = \frac{\delta l}{l}$$

Lateral Strain (ϵ'):- Strain measured lateral (transverse) to the linear strain is called lateral strain.

- For Round Bar

P = axial tensile load

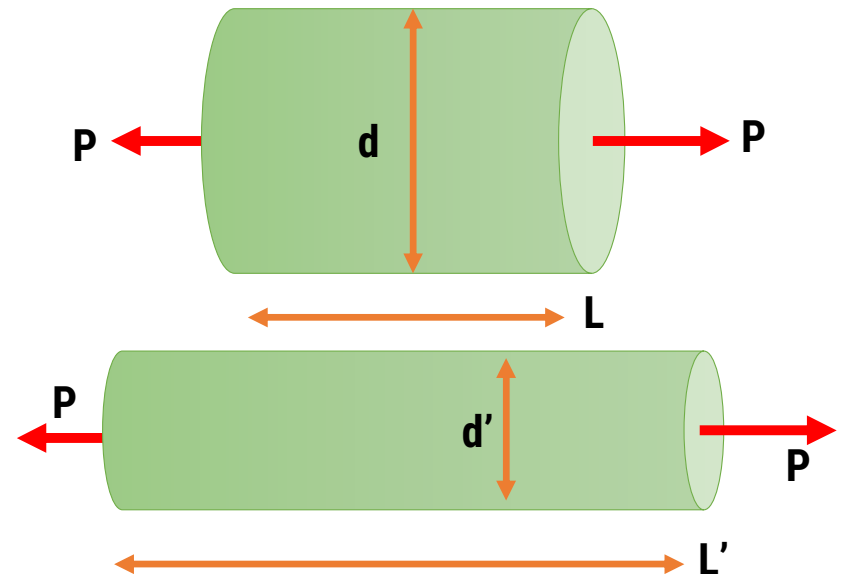
L = original length

L' = final length

D = original diameter

D' = final diameter

$$\text{Lateral strain}(\epsilon') = \frac{\text{Change in lateral dimension}}{\text{Original lateral dimension}} = \frac{\delta d}{d}$$



For Rectangle bar

$$\text{Lateral strain} = \frac{\text{Change in width}}{\text{Original width}} \quad \varepsilon' = \frac{\delta b}{b}$$

$$\text{Lateral strain} = \frac{\text{Change in thickness}}{\text{Original thickness}} \quad \varepsilon' = \frac{\delta t}{t}$$

Poisson's ratio (μ or $1/m$):- Ratio of lateral strain to linear strain is called poisson's ratio

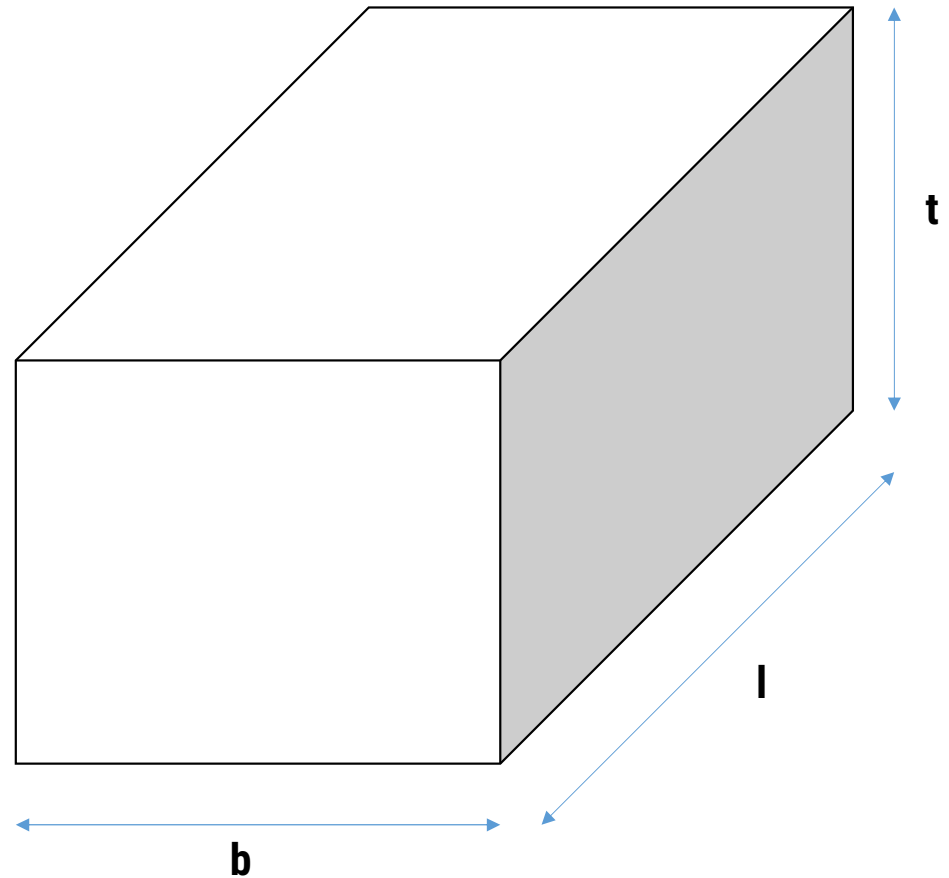
$$\text{Poisson's ratio}(\mu) = \frac{\text{Lateral strain}}{\text{Linear strain}} = \frac{\varepsilon'}{\varepsilon}$$

Volumetric strain or Dilatation (ε_v):- Ratio of change in volume to the original volume is called volumetric strain. It is also known as dilation.

$$\text{Volumetric strain}(\varepsilon_v) = \frac{\text{Change in volume}}{\text{Original volume}} = \frac{\delta V}{V}$$

Bulk modulus (K) :- When a body is subjected to three mutually perpendicular stresses of equal intensity, the ratio of direct stress to corresponding volumetric strain is known as bulk modulus.

$$\text{Bulk modulus (K)} = \frac{\text{Direct stress}}{\text{Volumetric strain}} = \frac{\sigma}{\varepsilon_v}$$



Volumetric strain of a rectangular body subjected to axial load:

Consider a bar, rectangular in section subjected to an axial tension force as shown in fig.

l = length of the bar, b = breadth of the bar, t = thickness of the bar

P = tensile force or load, E = modulus of elasticity, $1/m = \mu$ = poisson's ration

Now, Linear strain, $\sigma = \frac{P}{bt}$ (Tensile)

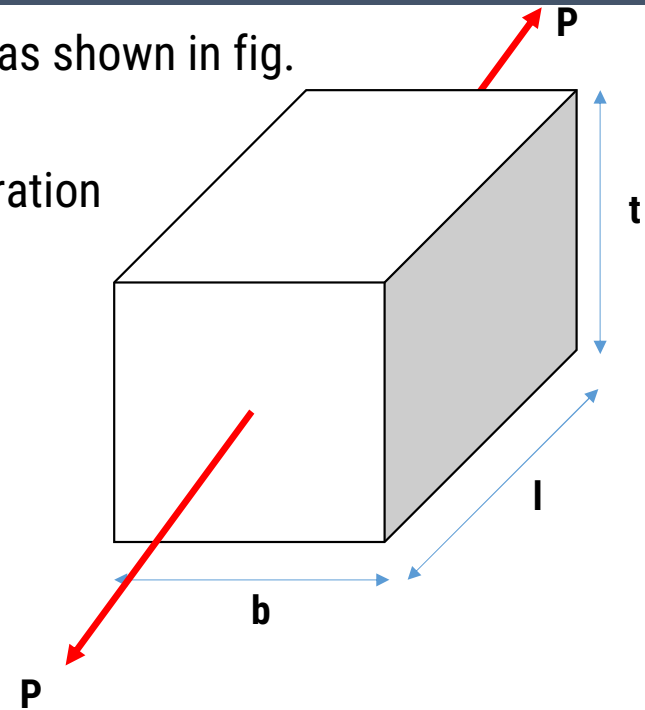
Longitudinal strain (ϵ_1), (Linear strain) (ϵ_1) $\epsilon_1 = \frac{\sigma}{E}$

Lateral strain (ϵ_2), $\epsilon_2 = -\frac{1}{m} \epsilon_1 = -\frac{\sigma}{mE}$

Here, $\frac{\delta l}{l} = \epsilon_1 = \frac{\sigma}{E}$

$$\frac{\delta b}{b} = \epsilon_2 = -\frac{\sigma}{mE}$$

$$\frac{\delta t}{t} = \epsilon_3 = -\frac{\sigma}{mE}$$



Volumetric strain of a rectangular body subjected to axial load:

So, Total volume of bar is $V = l \times b \times t$

Change in volume or volumetric strain

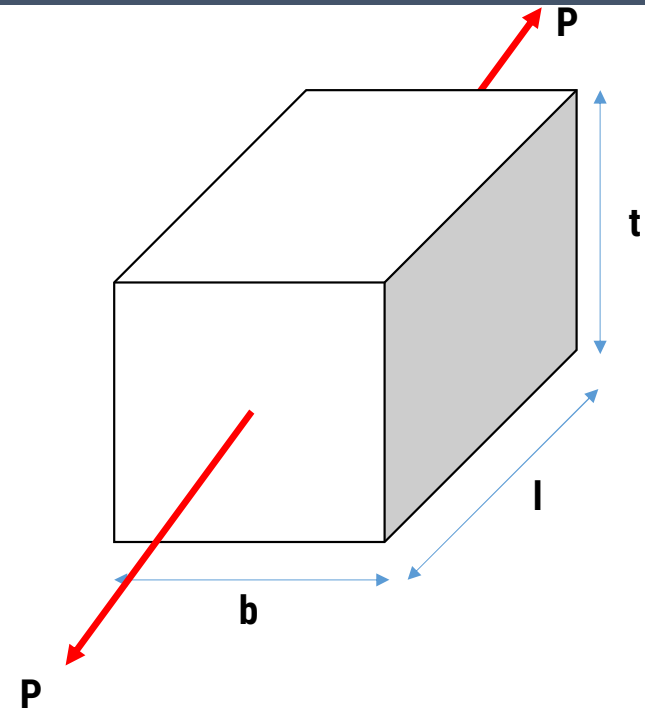
$$\frac{\delta V}{V} = \frac{\delta l}{l} + \frac{\delta b}{b} + \frac{\delta t}{t} = \varepsilon_v = \frac{\sigma}{E} - \frac{\sigma}{mE} - \frac{\sigma}{mE}$$

$$\therefore \varepsilon_v = \frac{\sigma}{E} \left(1 - \frac{1}{m} - \frac{1}{m}\right) = \frac{\sigma}{E} \left(1 - \frac{2}{m}\right)$$

$$\therefore \varepsilon_v = \frac{\sigma}{E} \left(1 - \frac{2}{m}\right) \quad \text{Where,} \quad \therefore \varepsilon_v = \frac{\sigma}{E} = \frac{\delta l}{l}$$

Change in volume $\delta V = V \times \frac{P}{btE} \left(1 - \frac{2}{m}\right)$

Volumetric strain $\varepsilon_v = \varepsilon(1 - 2\mu)$



Ex - 8

Determine the stress, strain, modulus of elasticity and Poisson's ratio from the following results for a bar tested on UTM : Diameter = 20 mm ; Gauge length = 150 mm ; increase in gauge length = 14 mm ; decrease in diameter = 0.85 mm ; Tensile load = 6 kN.

Given Data,

$$D_0 = 20 \text{ mm}$$

$$\delta L = 14 \text{ mm}$$

$$A_0 = 314.15 \text{ mm}^2$$

$$\delta d = 0.85 \text{ mm}$$

$$L_0 = 150 \text{ mm}$$

$$P = 6 \text{ kN} = 6 \times 10^3 \text{ N}$$

stress (σ)

$$\sigma = \frac{P}{A} = \frac{6 \times 1000}{314.15} = 19.10 \text{ N / mm}^2$$

strain (ε)

$$\varepsilon = \frac{\delta L}{L_0} = \frac{14}{150} = 0.0933$$

Lateral strain (ε')

$$\varepsilon' = \frac{\delta d}{d_0} = \frac{0.95}{20} = 0.0425$$

Poisson's ratio (μ)

$$\mu = \frac{\varepsilon'}{\varepsilon} = \frac{0.0425}{0.0933} = 0.4555$$

Modulus of elasticity (E)

$$E = \frac{\sigma}{\varepsilon} = \frac{19.10}{0.0933} = 204.71 \text{ N / mm}^2$$

Ex – 9:- Cement concrete block having a shape of square cross section of 250 mm side & a uniform height of 350 mm is tested in a compression testing machine by applying an axial compressive load of 'P'. It was observed that the height decreased by 0.28 mm & the side increased by 0.035 mm. If the modulus of elasticity of concrete is $0.213 \times 10^5 \text{ N/mm}^2$, determine (1) Poisson's ratio (2) Value of 'P' Load (3) Volumetric strain of the block.

Given Data,

$$A = 250 \times 250 = 62500 \text{ mm}^2$$

$$l = 350 \text{ mm}$$

$$b = 250 \text{ mm}$$

$$\delta b = 0.035 \text{ mm}$$

$$\delta l = 0.280 \text{ mm}$$

$$E = 0.130 \times 10^5 \text{ N/mm}^2$$

Linear Strain (ε)

$$\varepsilon = \frac{\delta l}{l_o} = \frac{0.280}{350} = 0.0008$$

Lateral strain (ε') $\varepsilon' = \frac{\delta b}{b_o} = \frac{0.035}{250} = 0.00014$

Poisson's ratio (μ) $\mu = \frac{\varepsilon'}{\varepsilon} = \frac{0.00014}{0.0008} = 0.175$

Stress develop in block (σ)

$$\sigma = E \times \varepsilon = 0.13 \times 10^5 \times 0.0008 = 10.4 \text{ N/mm}^2$$

Compressive force (P)

$$P = \sigma \times A = 10.4 \times 62500 = 650 \text{ kN}$$

Volumetric strain (ε_v)

$$\varepsilon_v = \frac{\delta V}{V} = \varepsilon(1 - 2\mu) = 0.00052$$

Volumetric strain of a rectangular body subjected to three mutually perpendicular forces

- Consider a rectangular body subjected to direct tensile stresses along three mutually perpendicular axes as shown in fig.

σ_X = Stress in x-x direction σ_Z = Stress in z-z direction

σ_Y = Stress in y-y direction E = Young modulus of elasticity

Now, strain in x-x direction due to stress σ_X $\epsilon_X = \frac{\sigma_X}{E}$

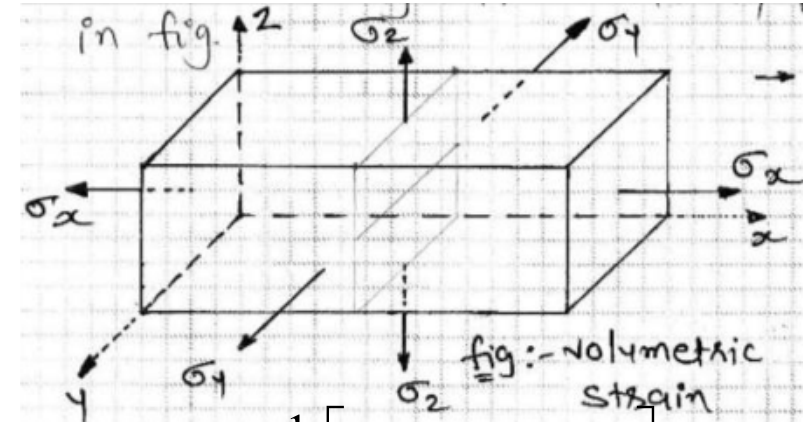
Similarly, $\epsilon_Y = \frac{\sigma_Y}{E}$ $\epsilon_Z = \frac{\sigma_Z}{E}$

The resulting strains in the three direction, may be found out by the principle of superposition, i.e. by adding algebraically the strain in each direction due to each individual stress.

Considering, tensile strain +ve & compression -ve.

Now, the resultant strain in x-direction,

$$\epsilon_X = \frac{\sigma_X}{E} - \frac{\sigma_Y}{mE} - \frac{\sigma_Z}{mE} = \frac{1}{E} \left[\sigma_X - \frac{\sigma_Y}{m} - \frac{\sigma_Z}{m} \right]$$



$$\epsilon_Y = \frac{\sigma_Y}{E} - \frac{\sigma_X}{mE} - \frac{\sigma_Z}{mE} = \frac{1}{E} \left[\sigma_Y - \frac{\sigma_X}{m} - \frac{\sigma_Z}{m} \right]$$

$$\epsilon_Z = \frac{\sigma_Z}{E} - \frac{\sigma_X}{mE} - \frac{\sigma_Y}{mE} = \frac{1}{E} \left[\sigma_Z - \frac{\sigma_X}{m} - \frac{\sigma_Y}{m} \right]$$

The volumetric strain may then be found by relation,

$$\epsilon_V = \frac{\delta V}{V} = \epsilon_X + \epsilon_Y + \epsilon_Z = (1 - 2\mu) \left(\frac{\sigma_X + \sigma_Y + \sigma_Z}{E} \right)$$

Relation between :- (1) Young's Modulus (E) (2) Bulk Modulus (K) (3) Poisson's ratio (μ)

► Consider a cube subjected to three mutually perpendicular tensile stresses of equal intensity

σ = Stress on the faces $V = l^3$

L = Length of cube, E = Young's modulus of cube

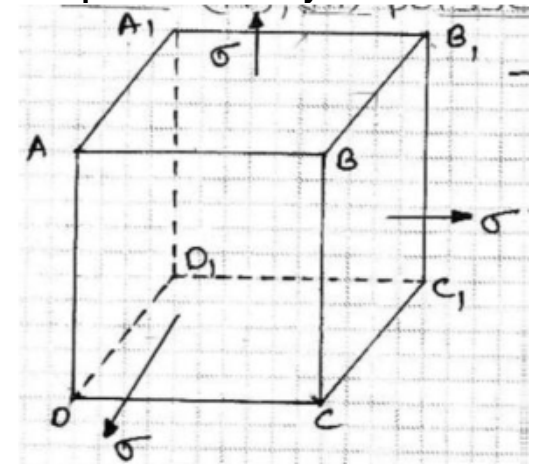
Consider the deformation of the side of the cube under the action of three mutually perpendicular tensile stress of equal intensity.

Now, Net tensile strain in side AB will be

- 1) Tensile strain equal to σ / E due to stresses on the face BB_1CC_1 & AA_1DD_1
- 2) Compressive lateral strain equal to σ / mE due to stresses on the face AA_1BB_1 & CC_1DD_1
- 3) Compressive lateral strain equal to σ / mE due to stresses on face $ABCD$ and $A_1B_1C_1D_1$

Now, Net tensile strain in side AB will be

$$\varepsilon = \frac{\delta l}{l} = \frac{\sigma}{E} - \left(\frac{1}{m} \times \frac{\sigma}{E} \right) - \left(\frac{1}{m} \times \frac{\sigma}{E} \right) = \frac{\sigma}{E} \left(1 - \frac{1}{m} \right) \dots\dots(1)$$



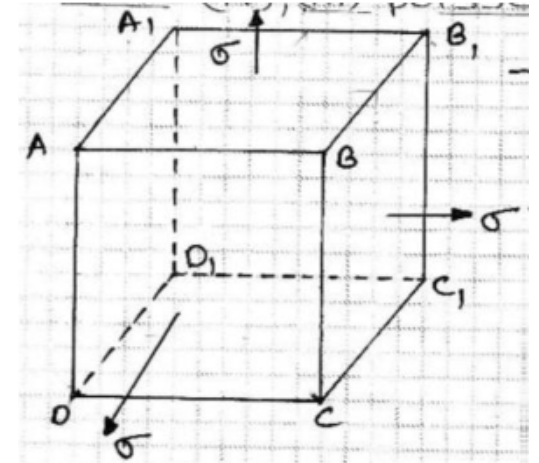
Relation between :- (1) Young's Modulus (E) (2) Bulk Modulus (K) (3) Poisson's ratio (μ)

Original volume of the cube

$$V = l^3 \quad (\text{Differentiate w.r.t } l) \quad \frac{\delta V}{\delta l} = 3l^2$$

$$\therefore \frac{\delta V}{V} = \frac{3l^2}{V} \delta l = 3 \frac{\delta l \times l^2}{l^3} = 3\varepsilon$$

From eqn (i) $\therefore \frac{\delta V}{V} = 3 \times \frac{\sigma}{E} \left(1 - \frac{2}{m} \right) \dots \dots \dots (2)$



Now, we know that

$$\text{Bulk modulus, } K = \frac{\sigma}{\varepsilon_v} = \frac{\sigma}{\delta V / V} = \frac{\sigma}{\frac{3\sigma}{E} \left(1 - \frac{2}{m} \right)} = \frac{E}{3 \left(1 - \frac{2}{m} \right)} = \frac{E}{3 \left(\frac{m-2}{m} \right)} = \frac{m}{3(m-2)}$$

$$E = 3K \left(1 - \frac{2}{m} \right) = 3K(1 - 2\mu)$$

Shear stress, shear strain & Modulus of rigidity or shear modulus

Shear Stress (τ) :- when a body is subjected to two equal & opposite forces acting tangentially across the resisting section, as a result of which the body tends to shear off across the section, the stress induced is called shear stress.

$$\text{shear stress} = \frac{\text{shear force}}{\text{shearing area}}$$

$$\tau = \frac{F}{A_s} (N / mm^2)$$

Shear Strain (ϕ) :-

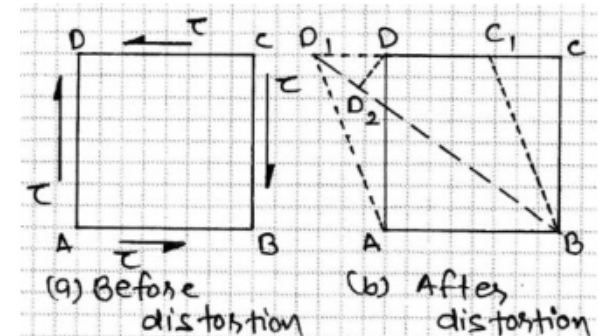
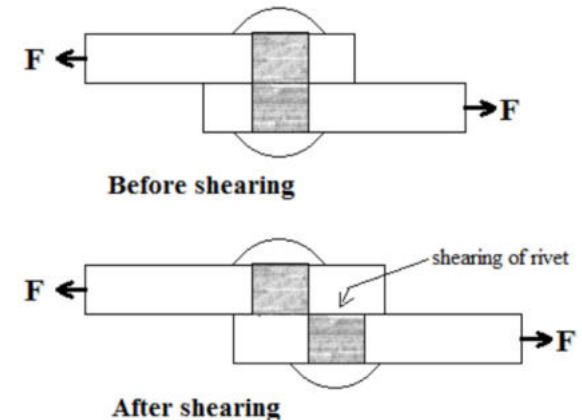
- Consider a cube of the length l fixed at bottom face AB
- Let force τ is applied to face DC tangentially to the face AB.
- As a result, the cube will deformed from ABCD to ABC₁ D₁.

$$\text{shear strain} = \frac{\text{deformation}}{\text{original length}} \Rightarrow \tan \phi = \phi = \frac{CC_1}{l}, \text{ but is } \phi \text{ every small}$$

Modulus of rigidity or & shear modulus:- (G, N or C)

$$\text{Modulus of rigidity} = \frac{\text{Shear Stress}}{\text{Shear Strain}}$$

$$C(\text{or } N \text{ or } G) = \frac{\tau}{\phi}$$



(1) Young's modulus (E), (2) Modulus of rigidity (G) & (3) Poisson's ratio (μ or $1/m$)

Consider a cube of length l , subjected to shear stress τ as shown in fig. due to shear stress diagonal BD will be elongated & diagonal AC will be shortened.

$$\text{Linear strain of } BD = \frac{BD_1 - BD}{BD} = \frac{D_1D_2}{BD} = \frac{DD_1 \cos 45^\circ}{AD\sqrt{2}} = \frac{DD_1}{AD}$$

$$\text{Here, } BD = \sqrt{AD^2 + AB^2} = \sqrt{AD^2 + AD^2} = \sqrt{2}AD$$

$$\therefore \text{Linear strain of } BD = \frac{\phi}{2} \text{ We know that, } G = \frac{\tau}{\phi} \Rightarrow \phi = \frac{\tau}{G}$$

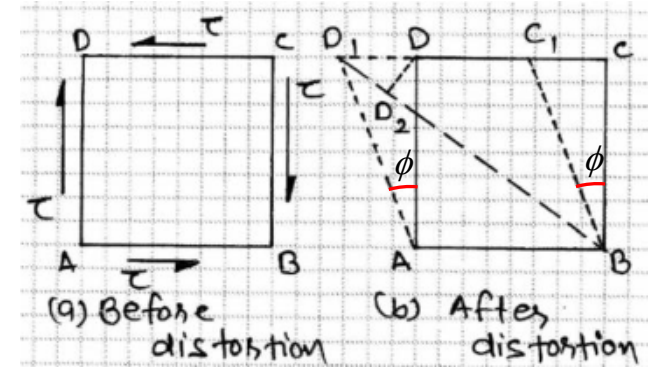
$$\therefore \text{Linear strain of } BD = \frac{\phi}{2} = \frac{\tau}{2G} \dots (1)$$

Now, tensile strain on diagonal BD due to tensile stress on diagonal

$$BD = \frac{\tau}{E} \dots (2)$$

& tensile strain on diagonal BD due to compressive stress on diagonal AC

$$= \frac{1}{m} \times \frac{\tau}{E} \dots (3)$$



Net strain on diagonal BD

$$= \frac{\tau}{E} + \frac{1}{m} \times \frac{\tau}{E} = \frac{\tau}{E} \left(1 + \frac{1}{m} \right) = \frac{\tau}{E} \left(\frac{m+1}{m} \right) \dots (4)$$

Now equating eqn (2) & (4)

$$\frac{\tau}{2G} = \frac{\tau}{E} \left(1 + \frac{1}{m} \right) \Rightarrow \therefore G = \frac{mE}{2(m+1)}$$

$$\therefore E = 2G \frac{m+1}{m} \Rightarrow \therefore E = 2G(1 + \mu)$$

(i) Young's modulus (E), (ii) Bulk modulus (K) & (iii) Modulus of rigidity (G)

We Know that, $K = \frac{mE}{3(m-2)}$

$$E = 3K \frac{(m-2)}{m} \Rightarrow E = 3K \left(1 - \frac{2}{m}\right)$$

$$E = 3K(1 - \mu) \Rightarrow \mu = \frac{\left(1 - \frac{E}{3K}\right)}{2} \dots\dots\dots(1)$$

Similarly, $G = \frac{mE}{2(m+1)}$

$$E = 2G \frac{(m+1)}{m} \Rightarrow E = 2G \left(1 + \frac{1}{m}\right)$$

$$E = 2G(1 + \mu) \Rightarrow \mu = \frac{E}{2G} - 1 \dots\dots\dots(2)$$

Equating equation (1) and (2)

$$\frac{\left(1 - \frac{E}{3K}\right)}{2} = \frac{E}{2G} - 1$$

$$\therefore 1 - \frac{E}{3K} = \frac{E}{G} - 2 \Rightarrow 3 = \frac{E}{G} + \frac{E}{3K} = E \left[\frac{1}{G} + \frac{1}{3K} \right] = E \left[\frac{3K + G}{3GK} \right] \Rightarrow E = \left[\frac{9GK}{G + 3K} \right]$$

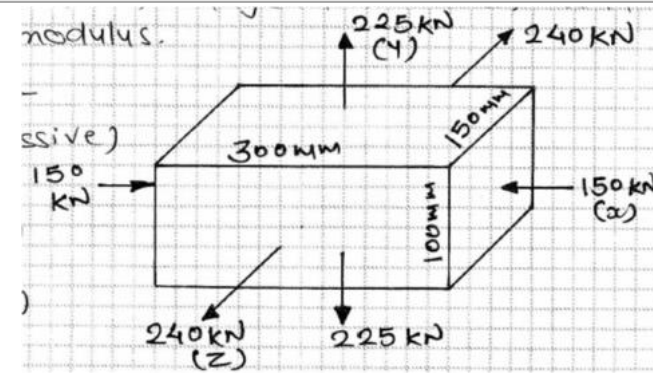
Ex – 10 :- A rectangular block of size 300mm x 150mm x 100mm is subjected to force of (i) compressive load of 150 kN in the direction its length (ii) tensile load of 225 kN on 300mm x 150mm (iii) tensile load of 240kN on the 300mm x 100mm face. If E=75 GPa & Poisson's ratio is 0.25, calculate (i) change in volume (ii) modulus of rigidity & (iii) bulk modulus.

Stresses on block:-

$$\sigma_x = \frac{150 \times 10^3}{150 \times 100} = 10 \frac{N}{mm^2} \quad \sigma_y = \frac{225 \times 10^3}{150 \times 300} = 5 \frac{N}{mm^2} \quad \sigma_z = \frac{240 \times 10^3}{100 \times 300} = 8 \frac{N}{mm^2}$$

(Compression) (Tensile) (Tensile)

$$E = 75 \text{ GPa} = 75 \times 10^3 \text{ N/mm}^2$$



Here, compressive force produce contraction of block, so compressive stress considered -ve(negative) & tensile stress considered +ve

$$\sigma_x = -10 \text{ N/mm}^2 \quad \sigma_y = 5 \text{ N/mm}^2 \quad \sigma_z = 8 \text{ N/mm}^2$$

Strain due to load:-

$$\epsilon_x = \frac{1}{E} [\sigma_x - \mu \sigma_y - \mu \sigma_z]$$

$$\epsilon_x = \frac{1}{E} [-10 - 0.25 \times 5 - 0.25 \times 8]$$

$$\epsilon_x = -\frac{13.25}{E}$$

$$\epsilon_y = \frac{1}{E} [\sigma_y - \mu \sigma_x - \mu \sigma_z]$$

$$\epsilon_y = \frac{1}{E} [-5 + 0.25 \times 10 - 0.25 \times 8]$$

$$\epsilon_y = -\frac{4.5}{E}$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \mu \sigma_x - \mu \sigma_y]$$

$$\epsilon_z = \frac{1}{E} [-8 + 0.25 \times 10 - 0.25 \times 5]$$

$$\epsilon_z = -\frac{6.75}{E}$$

Change in volume:-

$$\frac{\delta V}{V} = \varepsilon_x + \varepsilon_y + \varepsilon_z = \frac{-13.25}{E} + \frac{-4.5}{E} + \frac{-6.75}{E} = \frac{-24}{E}$$

$$\delta V = (150 \times 300 \times 1000) \times \frac{-24}{E} = -1440 \text{ mm}^3 \quad \text{decrease in volume}$$

Modulus of rigidity

$$G = \frac{mE}{2(m+1)} = \frac{4 \times 75 \times 10^3}{2(4+1)} = 30000 \text{ N / mm}^2$$

Bulk modulus

$$K = \frac{mE}{3(m-2)} = \frac{4 \times 75 \times 10^3}{3(4-2)} = 50000 \text{ N / mm}^2$$